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CZ3005 Artificial Intelligence

Week 11a – Default Logic

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Learning Goals

Understanding the:

- Default Rules
- Default Theory
- Default Logic Inference
 - Reiter Default Logic (RDL) inference
 - Limitations of RDL

Default Logic

- Standard logic can only express that something is true or that something is false.
 - **Explicitly given rules** (i.e., something is true if and only if it is explicitly stated as true in the knowledge base)
- Default logic is proposed to formalize reasoning with default assumptions.
 - **Common sense** (i.e., if there is nothing in the knowledge base that conflicts with our common sense, then we apply our common sense)
- It can express facts like
 - “by default, something is true”.

Default Logic



- What do we see here?
 - A child is using his handphone
- What do we intuitively think is going on?
 - He is playing a mobile game
- What do we intuitively conclude?
 - Doing so might negatively affect his study

Default Logic

- Default Rules
 - Knowledge Base
- 
- Default Theory
- Default Logic Inference
 - Reiter Default Logic Inference
 - Makinson Approach
 - Process Tree

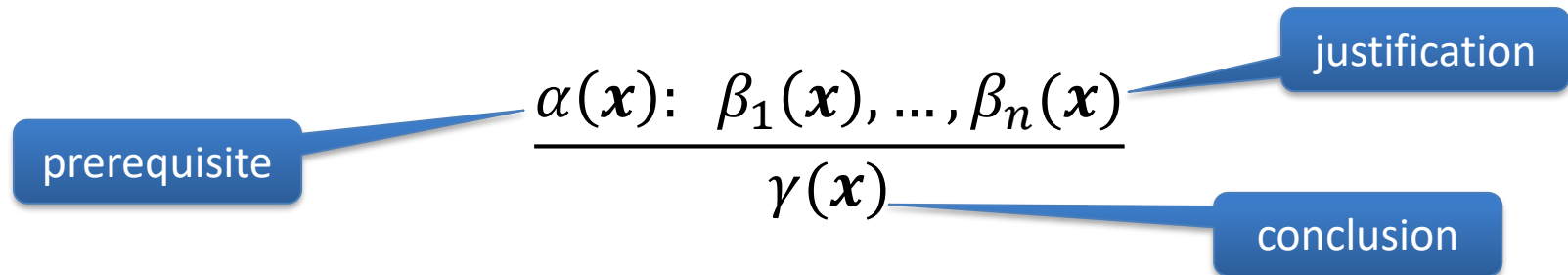
Default Rule

$$\frac{\text{Prerequisite : Justification}_1, \dots, \text{Justification}_n}{\text{Conclusion}}$$

According to this default:

- If we believe that Prerequisite is true;
- AND each of Justifications is consistent with our current beliefs;
- THEN, we are led to believe that the Conclusion is true.

Default Rule



where $\mathbf{x} = x_1, \dots, x_m$, and $\alpha(\mathbf{x}), \beta_1(\mathbf{x}), \dots, \beta_n(\mathbf{x}), \gamma(\mathbf{x})$ are formulae whose free variables are among $x_1, \dots, x_m$.

e.g.,:

$$\frac{bird(\mathbf{x}): has_wings(\mathbf{x})}{fly(\mathbf{x})}$$

“By default, a bird can fly, unless we know that a particular bird has lost its wings”

Default Rule

The default is **applied** by substituting c (the ground instance) into α and β to infer γ :

- Trigger: $\alpha(c)$ belongs to our set of beliefs.
- Justification: the set of our beliefs is consistent with each $\beta(c)$.

e.g.,:

Our KB does not
entail $\neg\beta(c)$

$$\frac{bird(x): has_wings(x)}{fly(x)} \quad \leftarrow \begin{matrix} \times \\ \{bird(Tweety), \neg has_wings(Tweety)\} \end{matrix}$$

Types of Default Rules

- Normal Defaults: $\frac{\alpha(x) : \gamma(x)}{\gamma(x)}$
- Semi-Normal Defaults: $\frac{\alpha(x) : \beta(x)}{\gamma(x)}$, where $\beta(x) \vdash \gamma(x)$

E.g., $\frac{bird(x) : has_wings(x)}{flies(x)},$

where $has_wings(x) \vdash flies(x)$

Types of Default Rules

- Open Defaults (Default Schemas) have unbounded variables, e.g., x

$$\frac{\alpha(x) : \beta_1(x), \dots, \beta_n(x)}{\gamma(x)}$$

- Closed (Grounded) Defaults use ground terms, e.g., $x=c$

$$\frac{\alpha(c) : \beta_1(c), \dots, \beta_n(c)}{\gamma(c)}$$

Default Theory

Delta: Set of Default Rules

$\langle \Delta, \Phi \rangle$

Phi: A given KB (a set of "Facts")

Example – the default rule that “birds typically fly”:

- $\Delta = \left\{ \frac{bird(x) : flies(x)}{flies(x)} \right\}$
 - This rule means that, "*if x is a bird, and there is no fact in the knowledge base suggesting that x cannot fly, then we can conclude that x flies*".
- $\Phi = \{bird(Tweety), cat(Sylvester)\}$

Example



- What do we see here?
 - A child is using his handphone (Pre-requisite)
- What do we intuitively think is going on?
 - He is playing a mobile game (Justification)
- What do we intuitively conclude?
 - Doing so might negatively affect his study (Conclusion)

Example



He is attending online class (not playing game)

- What do we see here?
 - A child is using his handphone (Pre-requisite)
- What do we intuitively think is going on?
 - He is playing a mobile game (Justification)
- What do we intuitively conclude?
 - Doing so might negatively affect his study (Conclusion)

Conflict with our default justification

Reiter Default Logic (RDL) Inference

- Guess the extension Ξ (pronounced as “Xi”)
- Initialise beliefs $\Xi^* = \Phi$
- (loop over) c-ground instance of an (unused) default $\frac{\alpha(x) : \beta(x)}{\gamma(x)}$:
 - Check two conditions
 - Triggered?: $\Xi^* \vdash \alpha(c)$
 - Justified?: $\beta(c)$ is consistent with Ξ
 - If yes: update beliefs $\Xi^* \leftarrow \Xi^* \cup \{\gamma(c)\}$
- (end loop)
- If $\Xi = \Xi^*$ then extension found/confirmed

The extension is added to our KB as
new knowledge

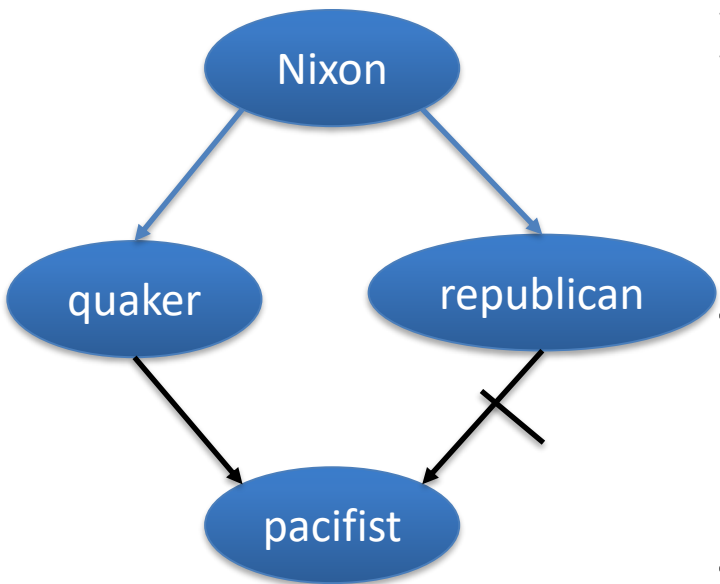
Example

Given Theory: $T = \left\langle \Delta = \left\{ \frac{bird(x) : flies(x)}{flies(x)} \right\}, \Phi = \{bird(Tweety), cat(Sylvester)\} \right\rangle,$

- Guess the extension = $Cn(\{flies(Tweety)\} \cup \Phi)$
- Our initial knowledge is $F = \Phi$
- Sylvester-instance of default not applicable:
 - not hold $\Phi \vdash bird(Sylvester)$
- $\Phi \vdash bird(Tweety)$ and $flies(Tweety)$ is consistent with F
- $F = \Phi \cup \{flies(Tweety)\}$
- No more default rules to apply
- An extension is reached

RDL Limitation - Nixon Diamond

The default rules may be applied in different orders, and this may lead to different extensions. E.g.:



Let Theory $T = \langle \Delta, \Phi \rangle$

$$\bullet = \left\{ \begin{array}{c} \frac{quaker(x) : pacifist(x)}{pacifist(x)}, \\ \frac{republican(x) : \neg pacifist(x)}{\neg pacifist(x)} \end{array} \right\}$$

$$\bullet \quad \Phi = \{ quaker(Nixon), republican(Nixon) \}$$

RDL Limitation - Nixon Diamond

Given $\langle \Delta, \Phi \rangle$

- $\Delta = \left\{ \frac{quaker(x) : pacifist(x)}{pacifist(x)}, \frac{republican(x) : \neg pacifist(x)}{\neg pacifist(x)} \right\}$
- $\Phi = \{quaker(Nixon), republican(Nixon)\}$

There are **two** extensions:

1. One that contains: **pacifist(Nixon)**
2. .. and the one that contains: **$\neg$ pacifist(Nixon)**

Conflicting conclusions

Addressing the RDL Limitation

- A default theory can have 0, 1 or more extensions.
- Entailment of a formula from a default theory can be defined in one of two ways:
 - Skeptical:
 - a formula is entailed by a default theory if it is entailed by all its extensions.
 - Credulous:
 - a formula is entailed by a default theory if it is entailed by at least one of its extensions.

Example

- The Nixon diamond example theory has two extensions:
 - one in which Nixon is a pacifist; and
 - one in which Nixon is not a pacifist.
- Thus, we have:
 - Neither $\text{Pacifist}(\text{Nixon})$ nor $\neg\text{Pacifist}(\text{Nixon})$ are skeptically entailed.
 - Both $\text{Pacifist}(\text{Nixon})$ and $\neg\text{Pacifist}(\text{Nixon})$ are credulously entailed.
- The credulous extensions of a default theory can be inconsistent with each other.

Thank you!

